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Sample plate supply control device, sample plate supply control system, sample plate supply control method and non-transitory computer readable medium storing sample plate supply control program

Abstract

A sample plate supply control device controls supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, and includes an identification information recognizer that recognizes identification information of a contained sample plate when a sample plate is contained in one sample plate container among the plurality of sample plate containers, and a supply instructor that instructs the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of a subject sample plate to be analyzed matches the identification information recognized by the identification information recognizer.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a sample plate supply control device including a sample plate holder that holds a sample plate to be supplied to an autosampler, and a sample plate supply control system. The present invention further relates to a sample plate supply control method and a non-transitory computer readable medium storing a sample plate supply control program.

BACKGROUND ART

(2) There is an analysis system including a liquid chromatograph. The analysis system includes a plurality of function units, a control device and a computer. The control device controls the plurality of function units. The computer examines chromatogram data. The plurality of function units have a pump, an oven, a detector and so on that are included in the liquid chromatograph. Further, the plurality of function units include an autosampler that injects a sample to be analyzed into the liquid chromatograph, a plate changer that supplies a sample plate containing a sample to the autosampler, etc. A program for controlling the analysis system and examining chromatogram data are installed, whereby the computer that examines chromatogram data functions as a CDS (Chromatography Data System).

(3) A user places a sample plate containing a sample vial in the plate changer in advance in order to start an analysis process. Next, the user operates the computer functioning as the CDS to select a sample plate for which the analysis process is to be started. In response to this operation, a subject sample plate is taken out from the plate changer and supplied to the autosampler. Thus, the autosampler injects a sample into the liquid chromatograph, and the analysis process is started.

(4) A plurality of sample plates can be contained in the plate changer. In a case where 30 sample plates can be contained in the plate changer, for example, the numbers from 1 to 30 are displayed on the plate changer to correspond to sample plate containers. When operating the computer to start the analysis process, the user designates the number of a sample plate container using the computer. When placing a sample plate in the sample plate container numbered 10, for example, the user provides an instruction for starting the analysis process with respect to the sample plate container numbered 10 using the computer.

(5) The below-mentioned Patent Document 1 discloses an analysis system comprising a rack waiting place (plate changer) that can contain a plurality of racks (sample plates).

(6) [Patent Document 1] JP 2006-189362 A

SUMMARY OF INVENTION

Technical Problem

(7) As described above, the plate changer that can contain the plurality of sample plates may be attached to the autosampler. The plate changer can contain the plurality of sample plates at the same time. Therefore, the user may erroneously select a position at which a sample plate is to be contained in the plate changer. Alternatively, the user may erroneously designate a position at which a sample plate is contained when operating the computer.

(8) The user may have an analysis instruction, etc. that may designate a position at which a sample plate is to be contained in advance. Even in such a case, in an environment in which a plurality of users use a common analysis system, another user may have already placed another sample plate at

a position, where a sample plate is to be contained, designated by the analysis instruction. Under such a circumstance, the user must replace the sample plate with the other sample plate.

(9) In a case where another user has already placed another sample plate at a position, where a sample plate is to be contained, designated by the analysis instruction, the user may place the sample plate in another available sample plate container. In this case, the user must operate the computer to change a set plate position.

(10) Further, after placing a sample plate in the plate changer, the user must move to the location of the computer and again operate the computer to provide an instruction for starting an analysis. In a case where the plate changer and the computer are located at a distance from each other, work is complicated and requires an unnecessary time.

(11) An object of the present invention is optimization of work for placing a sample plate and provision of an instruction for supplying the sample plate in a sample plate supply control device including a sample plate holder that can contain a plurality of sample plates, and a sample plate supply control system.

Solution to Problem

(12) (1) A sample plate supply control device according to one aspect of the present invention controls supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, and includes an identification information recognizer that recognizes identification information of a contained sample plate when a sample plate is contained in one sample plate container among the plurality of sample plate containers, and a supply instructor that instructs the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of a subject sample plate to be analyzed matches the identification information recognized by the identification information recognizer.

(13) This sample plate supply control device recognizes the identification information of the sample plate contained in the one sample plate container among the plurality of sample plate containers. Then, in a case where the identification information of the subject sample plate matches the recognized identification information of the sample plate, the sample plate supply control device supplies the sample plate contained in the one sample plate container to the autosampler. Thus, even in a case where the sample plate is contained in any sample plate container, the subject sample plate is supplied to the autosampler.

(14) The user does not have to be conscious of which sample plate container the sample plate is to be contained in, in the sample plate holder. The user may place the sample plate in an available sample plate container. An analysis process does not start with respect to a different sample plate due to an erroneous operation. Even in a case where a plurality of users use a common sample plate holder, each user can freely utilize an available sample plate container.

(15) (2) The supply instructor may notify the autosampler of process information and may provide an instruction for executing a process on the sample plate contained in the one sample plate container based on the process information in a case where the identification information of the subject sample plate matches the identification information recognized by the identification information recognizer.

(16) In a case where the identification information of the subject sample plate matches the recognized identification information of the sample plate, this sample plate supply control device supplies the sample plate contained in the one sample plate container to the autosampler. At this time, the sample plate supply control device notifies the autosampler of process information in regard to the subject sample plate. Thus, even in a case where the sample plate is contained in any sample plate container, the process is started with respect to the subject sample plate based on the process information.

(17) (3) The supply instructor may wait until a process with respect to another sample plate is completed and may instruct the sample plate holder to supply the sample plate contained in the one

sample plate container to the autosampler in a case where the identification information of the subject sample plate matches the identification information recognized by the identification information recognizer, and the process is being executed on the another sample plate in the autosampler.

(18) In a case where the identification information of the subject sample plate matches the recognized identification information of the sample plate, this sample plate supply control device determines whether the autosampler is executing a process with respect to another sample plate. When the autosampler is executing a process with respect to another sample plate, the sample plate supply control device supplies the sample plate contained in the one sample plate container to the autosampler after the process with respect to the other sample plate is completed. Thus, even when the autosampler is executing a process with respect to the other sample plate, reservation of execution of a process with respect to the subject sample plate is made. The user can make settings for the subject sample plate regardless of whether the autosampler is executing a process with respect to another sample plate.

(19) (4) The supply instructor may instruct the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of any subject sample plate among a plurality of subject sample plates matches the identification information recognized by the identification information recognizer.

(20) In a case where any identification information of the plurality of subject sample plates matches the recognized identification information of the sample plate, this sample plate supply control device supplies the sample plate contained in the one sample plate container to the autosampler. Thus, even in a case where the sample plate is contained in any sample plate container, any subject sample plate among the plurality of subject sample plates is supplied to the autosampler. The user can make reservation for the analysis process with respect to the plurality of subject sample plates at the same time.

(21) (5) A print module that prints the identification information of the subject sample plate to be attached to the subject sample plate may further be included.

(22) The identification information to be attached to the subject sample plate is printed. The user can make settings for the subject sample plate and print the identification information to be attached to the subject sample plate.

(23) (6) The identification information of the subject sample plate may be attached to the subject sample plate as a barcode image.

(24) The identification information is attached to the subject sample plate as a barcode image. The identification information of the subject sample plate can be read with use of a barcode reading function.

(25) (7) A sample plate supply control system according to another aspect of the present invention includes an autosampler that injects a sample to be analyzed into an analysis device, a sample plate holder that has a plurality of sample plate containers, and a terminal device that controls the sample plate holder, wherein the sample plate holder includes an identification information recognizer that recognizes identification information of a contained sample plate when the sample plate is contained in one sample plate container among the plurality of sample plate containers, and an identification information presenter that supplies the identification information of the contained sample plate to the terminal device, and the terminal device includes an identification information acquirer that acquires the identification information of the contained sample plate from the sample plate holder, and a supply instructor that instructs the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of a subject sample plate to be analyzed matches the identification information acquired by the identification information acquirer.

(26) This sample plate supply system recognizes the identification information of the sample plate contained in the one sample plate container among the plurality of sample plate containers. Then,

in a case where the identification information of the subject sample plate matches the recognized identification information of the sample plate, the sample plate supply control system supplies the sample plate contained in the one sample plate container to the autosampler. Thus, even in a case where the sample plate is contained in any sample plate container, the subject sample plate is supplied to the autosampler.

(27) (8) The sample plate holder may have a plurality of displays provided to correspond to the plurality of sample plate containers, and each display may display a processing state in regard to each sample plate contained in each sample plate container.

(28) The display displays a processing state of a sample plate in each sample plate container. The user can identify an available sample plate container.

(29) (9) A sample plate supply control method according to yet another aspect of the present invention of controlling supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, includes acquiring identification information of a contained sample plate from the sample plate holder when the sample plate is contained in one sample plate container among the plurality of sample plate containers, and instructing the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of a subject sample plate to be analyzed matches the identification information acquired from the sample plate holder.

(30) (10) A non-transitory computer readable medium storing a sample plate supply control program according to yet another aspect of the present invention controls supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, wherein the sample plate supply control program causes a computer to execute the processes of acquiring identification information of a contained sample plate from the sample plate holder when the sample plate is contained in one sample plate container among the plurality of sample plate containers, and instructing the sample plate holder to supply the sample plate contained in the one sample plate container to the autosampler in a case where identification information of a subject sample plate to be analyzed matches the identification information acquired from the sample plate holder.

(31) (11) The sample plate supply control device described in (1) may further include a subject plate setter that sets the identification information of the subject sample plate and a method file in which an execution condition of an analysis process is written, in association with each other, wherein the supply instructor may have a process instructor that acquires a method file associated with the subject sample plate by the subject plate setter and instructs the autosampler to execute the analysis process based on the method file.

Advantageous Effects of Invention

(32) The present invention can optimize work for placing a sample plate and provision of an instruction for supplying the sample plate in a sample plate supply control device including a sample plate holder that can contain a plurality of sample plates and a sample plate supply control system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an overview of an analysis system according to the present embodiment.

(2) FIG. 2A is a plan view of a sample plate, and FIG. 2B is a back view of a sample plate.

(3) FIG. 3 is a diagram showing a plate changer and an autosampler according to the present embodiment.

(4) FIG. 4 is a block diagram of the functions of a sample plate supply control system according to the present embodiment.

(5) FIG. 5 is a diagram showing a plate setting screen displayed in a monitor of a computer according to the present embodiment.

(6) FIG. 6 is a flowchart of a process executed in the computer according to the present embodiment.

(7) FIG. 7 is a flowchart of the process executed in the computer according to the present embodiment.

(8) FIG. 8 is a flowchart of the process executed in the computer according to the present embodiment.

DESCRIPTION OF EMBODIMENTS

(9) A sample plate supply control device, a sample plate supply control system, a sample plate supply control method and a sample plate supply control program according to embodiments of the present invention will be described below in detail with reference to the drawings.

[1] First Embodiment

(10) (1) Overall Configuration of Analysis System

(11) FIG. 1 is an overview of an analysis system **10** according to the present embodiment. The analysis system **10** shown in FIG. 1 includes a computer **1**, a network **2**, a server **3**, a control device **4**, a plurality of function units **5** and a printing machine **9**. In the present embodiment, a liquid chromatograph **50** is constituted by the plurality of function units **5**. The liquid chromatograph **50** is a high-performance liquid chromatograph, for example.

(12) The computer **1** is a general personal computer, for example. The computer **1** executes a process of controlling the analysis system **10**, a process of examining chromatogram data, etc.

(13) The network **2** is a computer network such as a LAN (Local Area Network). The computer **1** and the server **3** are connected to each other via the network **2**.

(14) The server **3** is a general computer server, for example. A batch table data base **30** (hereinafter abbreviated as a batch table DB **30**) is saved in a storage of the server **3**. The batch table DB **30** accumulates a plurality of batch tables. The batch table is the data in which the execution procedure and execution conditions of an analysis process to be executed in the analysis system **10** are written. The batch table is set in regard to a sample plate **8** (FIGS. 2A and 2B) that contains a plurality of sample vials.

(15) The batch table includes a plurality of records. One record is a record corresponding to one sample vial contained in a sample plate. In one record, the identification information of a sample plate **8**, the number of a sample vial, a sample type and the name of a method file are written. A method file is a data file in which the execution procedure and execution conditions of the analysis process are written. That is, in the batch table, a method file in which the content of an analysis process is written is set for each sample vial. Because a plurality of sample vials are contained in the sample plate **8**, the records corresponding to all of the sample vials contained in the sample plate **8** are included in the batch table.

(16) The control device **4** is connected to the computer **1** via a communication cable. Further, the control device **4** is connected to the plurality of function units **5** via communication cables. The analysis system **10** includes the plurality of function units **5**. The plurality of function units **5** include a pump, an oven, a detector and so on in the liquid chromatograph **50**. Further, the plurality of function units **5** include an autosampler **7** (see FIG. 3) that injects a sample to be analyzed into the liquid chromatograph **50**, a plate changer **6** (see FIG. 3) that supplies a sample plate containing a sample to the autosampler **7**, etc.

(17) (2) Configuration of Sample Plate and Plate Changer

(18) FIG. 2A shows a plan view of the sample plate **8**. FIG. 2B shows a back view of the sample plate **8**. FIG. 3 is a diagram showing the configuration of the plate changer **6**. First, the sample plate **8** will be described with reference to FIGS. 2A and 2B.

(19) As shown in FIG. 2A, a plurality of holes **81** in which sample vials are contained are provided in the sample plate **8**. As shown in FIG. 2B, a barcode **91** is attached to the back surface of the

sample plate **8** of the present embodiment. The information for identifying the sample plate **8** is written in the barcode **91** as identification information. Although details will be described below, the barcode **91** is read when the sample plate **8** is contained in the plate changer **6**, so that the contained sample plate **8** is specified.

(20) As shown in FIG. 3, the plate changer **6** is provided next to the autosampler **7** as a device attached to the autosampler **7**. The plate changer **6** can hold a plurality of sample plates **8**. The sample plate **8** can contain a plurality of sample vials containing samples to be analyzed.

(21) In the plate changer **6** of the present embodiment, three changers **6A**, **6B**, **6C** are provided. Each of the three changers **6A**, **6B**, **6C** can contain a plurality of sample plates **8** (fourteen sample plates **8** can be contained in this example). The changer **6A** includes sample plate containers **61** numbered 3 to 16. The changer **6B** includes sample plate containers **61** numbered 17 to 30. The changer **6C** includes sample plate containers **61** numbered 31 to 44. Since sample plate containers numbered 1 and 2 are provided in the autosampler **7**, numbering of the sample plate containers **61** in the changer **6A** starts from 3.

(22) As shown in FIG. 3, a display **62** is provided next to each sample plate container **61**. An LED is used in a display **62** in the present embodiment. A display **62** displays a processing state of a sample plate **8** contained in a sample plate container **61**. In the present embodiment, in a case where an analysis process of a sample plate **8** contained in a sample plate container **61** is not completed, a display **62** lights up in red. In a case where a display **62** next to a sample plate container **61** numbered 25 lights up in red, for example, the user can judge that a sample plate **8** contained in the sample plate container **61** numbered 25 has not been processed. That is, the user can judge that the sample plate **8** contained in the sample plate container **61** numbered 25 is in a standby state.

(23) While a display **62** lights up in red in a case where an unprocessed sample plate **8** is contained in the present embodiment, another processing state of the sample plate **8** may be identified. For example, in regard to a sample plate container **61** containing a sample plate **8** for which an analysis process is completed, an adjacent display **62** may light up in green. Alternatively, in a case where a sample plate **8** is being transported and carried into the autosampler **7**, that is, in regard to a sample plate container **61** corresponding to a sample plate for which an analysis process is being executed, an adjacent display **62** may light up in orange.

(24) (3) Functional Configuration of Sample Plate Supply Control System

(25) FIG. 4 is a block diagram of the functions of the sample plate supply control system **20**. As shown in FIG. 4, the plate changer **6** includes a reader **63**, a plate transport device **64**, an identification information recognizer **65** and an identification information presenter **66**. The plate changer **6** has a controller **60** constituted by a CPU (Central Processing Unit), a RAM (Random Access Memory), a ROM (Read Only Memory) and so on. A computer program such as a first control program is stored in the ROM. The CPU executes the first control program stored in the ROM on the RAM, whereby functions of the identification information recognizer **65** and the identification information presenter **66** are implemented. Part or all of the identification information recognizer **65** and the identification information presenter **66** may be implemented by hardware such as an electronic circuit.

(26) The reader **63** acquires the information written in the barcode **91** by reading the barcode **91** attached to a sample plate **8** contained in each sample plate container **61**, and supplies the acquired information and the number of a sample plate container **61** storing the sample plate **8** to the identification information recognizer **65**.

(27) The plate transport device **64** takes out a sample plate **8** contained in any one of a plurality of sample plate containers **61**, and supplies the taken sample plate **8** to the autosampler **7**. The plate transport device **64** receives the sample plate **8** from the autosampler **7** after an analysis process is completed, and places the received sample plate **8** in an original sample plate container **61**.

(28) In the present embodiment, a sample plate **8** contained in each sample plate container **61** is

taken out once from the sample plate container **61** by the plate transport device **64** for reading of a barcode **91** by the reader **63**. The plate transport device **64** moves the sample plate **8** taken out from the sample plate container **61** to a position where the reader **63** can perform an operation of reading the barcode **91**. When the operation of reading the barcode **91** with respect to the sample plate **8** is completed, the plate transport device **64** transports the sample plate **8** to the original sample plate container **61**. A reader **63** may be provided individually for each sample plate container **61**.

(29) The identification information recognizer **65** recognizes identification information of a sample plate **8** contained in a sample plate container **61**. Specifically, the identification information recognizer **65** acquires the identification information of a contained sample plate **8** from the information received from the reader **63**. Then, the identification information recognizer **65** supplies the recognized identification information of the sample plate **8** and the number of the sample plate container **61** containing the sample plate **8** to the identification information presenter **66**.

(30) The identification information presenter **66** transmits the identification information of the sample plate **8** received from the identification information recognizer **65** and the number of the sample plate container containing the sample plate **8** to the control device **4**. The control device **4** transmits the identification information of the sample plate **8** and the number of the sample plate container **61** containing the sample plate **8**, which are received from the identification information presenter **66**, to the computer **1** via the control device **4**.

(31) The computer **1** includes a subject plate setter **11**, a subject plate storage **12**, an identification information acquirer **13**, a supply instructor **14**, a print module **15**, a monitor **18** and an operation unit **19**. The supply instructor **14** has a process instructor **141**. The computer **1** is constituted by a CPU (Central Processing Unit), a RAM (Random Access Memory), a ROM (Read Only Memory) and a storage device such as a hard disc. A computer program such as a second control program is stored in the ROM. The CPU executes the second control program stored in the ROM or the storage device on the RAM, whereby functions of the subject plate setter **11**, the identification information acquirer **13**, the supply instructor **14** and the print module **15** are implemented. Part or all of these functions **10** may be implemented by hardware such as an electronic circuit. Further, an examination program for examining chromatogram data is installed in the computer **1**. In this manner, the computer **1** functions as a CDS (Chromatography data system) using the second control program and the examination program. The subject plate storage **12** is constituted by a storage device such as a RAM or a hard disc.

(32) The subject plate setter **11** makes settings of an analysis process with respect to a sample plate **8** to be analyzed. A sample plate **8** to be analyzed is referred to as a subject sample plate **8T** for the sake of explanation in the following description. The subject plate setter **11** sets plate identification information of a subject sample plate **8T**, the number of a sample vial, a sample type and the name of a method file in association with one another as one record. The subject plate setter **11** can set a plurality of records in which plate identification information pieces of subject sample plates **8T**, the numbers of sample vials, sample types and the names of method files are respectively associated with one another. The subject plate setter **11** puts a plurality of records having the same plate identification information of a subject sample plate **8T** into a group and sets the group as a batch table.

(33) The subject plate storage **12** stores a batch table that is set with respect to subject sample plates **8T** in the subject plate setter **11**.

(34) The identification information acquirer **13** receives identification information of a sample plate **8** and the number of a sample plate container **61** containing the sample plate **8**, which have been transmitted by the identification information presenter **66** via the control device **4**.

(35) The supply instructor **14** compares the identification information of a subject sample plate **8T** stored in the subject plate storage **12** with the identification information of a sample plate **8** acquired by the identification information acquirer **13**. In a case where the identification

information of the subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the supply instructor **14** determines that the sample plate **8** contained in the sample plate container **61** is a subject sample plate **8T** to be processed. In a case where determining that the subject sample plate **8T** is contained in the sample plate container **61**, the supply instructor **14** provides an instruction for supplying the sample plate by the sample plate supply control system **20**. The content of a process of providing an instruction for supplying a sample plate will be described below in detail.

(36) The print module **15** generates a barcode **91** to be attached to a subject sample plate **8T** and provides an instruction for printing. The print module **15** instructs the printer **9** to print the barcode **91**.

(37) (4) Behavior of Setting Plate by Computer

(38) Next, a plate setting operation of setting a subject sample plate **8** will be described. The plate setting operation is performed by the subject plate setter **11** included in the computer **1**. FIG. **5** is a diagram showing a plate setting screen **100** displayed in the monitor **18** of the computer **1** when the subject plate setter **11** is executed.

(39) The plate setting screen **100** includes a plate identification information field **101** in which identification information of a subject sample plate **8T** is input, a batch table name field **102** for designation of the name of a batch table, a barcode print instruction button **103** and a standby instruction button **104**.

(40) In the plate identification information field **101**, the identification information for specifying a subject sample plate **8T** is input by the user. The user operates the operation unit **19** of the computer **1** and inputs the identification information for specifying a subject sample plate **8T** using characters, numerals, symbols or a combination of these. The user may set any identification information to be input. Alternatively, the identification information to be input is set in accordance with an analysis manual or the like.

(41) In the batch table name field **102**, the name of a batch table selected by the user is input. In the present embodiment, the subject plate setter **11** acquires a batch table from the batch table DB **30** stored in the server **3** shown in FIG. **1**. The user operates the operation unit **19** and can cause a screen showing a list of batch tables to be displayed by adjusting a pointer to the batch table name field **102**. The subject plate setter **11** accesses the batch table DB **30** of the server **3**, acquires a prepared list of a plurality of batch tables and causes the monitor **18** to display the list in the screen.

(42) In the present embodiment, the subject plate setter **11** accesses the batch table DB **30** of the server **3**, whereby the user can set a batch table. That is, the user can select a prepared batch table.

For example, the user may be able to create a batch table in the plate setting screen **100**.

Alternatively, the user may be able to edit a batch table, which the user has acquired from the server **3**, in the plate setting screen **100**.

(43) The barcode print instruction button **103** generates a barcode **91** and provides an instruction for printing the barcode **91**. In a case where the user operates the operation unit **19** to select the barcode print instruction button **103**, the print module **15** generates image data of the barcode **91**. The print module **15** converts the identification information input in the plate identification information field **101** into a barcode and generates image data of the barcode **91**. The print module **15** transmits an instruction for printing the image data of the generated barcode **91** to the printer **9**.

(44) The printer **9** generates a sticker on which the image of the barcode **91** is printed. As shown in FIGS. **2A** and **2B**, the user attaches the sticker on which the barcode **91** is printed to a predetermined position on the back surface of a sample plate **8**. In the present embodiment, the reader **63** performs a reading operation with respect to the back surface of a sample plate **8**, so that the sticker of the barcode **91** is attached to the back surface of the sample plate **8**. Depending on the configuration of the reader **63**, the sticker of the barcode **91** may be attached to the side surface or the upper surface of a sample plate **8**.

(45) With the above-mentioned operation, a batch table corresponding to a subject sample plate **8T** is set. Further, the identification information of a subject sample plate **8T** is converted into a barcode, and a sticker of a barcode **91** is printed. As shown in FIG. **2B**, the user attaches the sticker of the barcode **91** to a predetermined position on the back surface of a subject sample plate **8T**.

(46) Reference is made to FIG. **5** again. Next, the user operates the operation unit **19** to select the standby instruction button **104**. In response to this operation, the subject plate setter **11** stores the information set in the plate setting screen **100** in the subject plate storage **12**. The subject plate storage **12** stores the identification information of a subject sample plate **8T** and a batch table.

(47) The user selects the standby instruction button **104** and then moves to the location of the plate changer **6**. The user places the subject sample plate **8T** in any sample plate container **61** in the plate changer **6**. At this time, the user may place the subject sample plate **8T** in any sample plate container **61**. While a large number of sample plate containers **61** are prepared as shown in FIG. **3** in the present embodiment, the user may place the subject sample plate **8T** in any sample plate container **61**.

(48) Preparation for an analysis process is completed by the above-mentioned operations and processes. That is, the user inputs the identification information of the subject sample plate **8T** in the plate setting screen **100**. The user sets the batch table in the plate setting screen **100**. The user prints the sticker of a barcode **91** and attaches the sticker to the subject sample plate **8T**. The user selects the standby instruction button **104**, whereby the subject plate storage **12** stores the identification information of the subject sample plate **8T** and the batch table. Then, the user places the subject sample plate to which the sticker of the barcode **91** is attached in any sample plate container **61**. Next, a process of supplying a sample plate **8** starts.

(49) (5) Behavior of Supplying Sample Plate by Plate Changer and Computer

(50) When the user places a subject sample plate **8T** in any sample plate container **61**, an operation of reading a barcode **91** is performed by the reader **63** in the plate changer **6**. The reader **63** supplies the information of the barcode **91** and the number of the sample plate container **61** containing the sample plate **8** to the identification information recognizer **65**. Since whether the sample plate **8** is a subject sample plate **8T** has not been recognized in the subsequent process, the sample plate **8** is not referred to as a subject sample plate **8T** but simply referred to as a sample plate **8**. The identification information recognizer **65** recognizes the identification information of the contained sample plate **8** from the received information of the barcode **91**. Then, the identification information recognizer **65** supplies the recognized identification information of the sample plate **8** and the number of the sample plate container **61** containing the sample plate **8** to the identification information presenter **66**.

(51) The identification information presenter **66** transmits the identification information of the sample plate **8** and the number of the sample plate container **61** to the control device **4**. The control device **4** transmits the identification information of the sample plate **8** and the number of the sample plate container **61** to the computer **1**. Next, the behavior of providing an instruction for supplying the sample plate **8** by the computer **1** is described with reference to the flowchart of FIG. **6**.

(52) FIG. **6** is a flowchart showing a process executed by the identification information acquirer **13** and the supply instructor **14** of the computer **1**. First, the identification information acquirer **13** of the computer **1** checks whether the identification information of the sample plate **8** and the number of the sample plate container **61** have been acquired (step **S101**). In a case where the identification information acquirer **13** acquires the identification information of the sample plate **8** and the number of the sample plate container **61**, the supply instructor **14** makes reference to the subject plate storage **12** (step **S102**). In a case where the identification information acquirer **13** does not acquire the identification information of the sample plate **8** and the number of the sample plate container **61**, the process ends.

(53) After the step **S102**, the supply instructor **14** determines whether the identification information

of a subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13** (step **S103**). In a case where the identification information of the subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the supply instructor **14** determines that the sample plate **8** contained in the sample plate container **61** is a subject sample plate **8T** to be processed. In a case where determining that the subject sample plate **8T** is contained in the sample plate container **61**, the supply instructor **14** instructs the plate changer **6** to supply the subject sample plate **8T**. That is, the supply instructor **14** instructs the plate changer **6** to supply the sample plate **8** having the identification information acquired by the identification information acquirer **13** to the autosampler **7** (step **S104**). Alternatively, the supply instructor **14** instructs the plate changer **6** to supply the sample plate **8** contained in the sample plate container **61** having the number acquired by the identification information acquirer **13** to the autosampler **7**. In a case where the identification information of the subject sample plate **8T** stored in the subject plate storage **12** does not match the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the supply instructor **14** ends the process.

(54) After the step **S104**, the process instructor **141** makes reference to the subject plate storage **12** and acquires a batch table corresponding to the subject sample plate **8T**. Then, the process instructor **141** notifies the autosampler **7** of the information relating to an analysis process based on the batch table (step **S105**). With the process mentioned above, the plate changer **6** supplies the subject sample plate **8T** to the autosampler **7**. Then, a sample is extracted from a sample vial contained in the subject sample plate **8T** by the autosampler **7**, and the sample is injected into the analysis device.

(55) As described above, the sample plate supply control system **20** of the present embodiment acquires the identification information of a sample plate **8** contained in any sample plate container **61** among the plurality of sample plate containers **61**. Then, in a case where the identification information of a subject sample plate **8T** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the sample plate supply control system **20** supplies the sample plate **8** the identification information of which has been acquired to the autosampler **7**. Thus, even in a case where the sample plate **8** is contained in any sample plate container **61**, the subject sample plate **8T** stored in the subject plate storage **12** is supplied to the autosampler **7**.

(56) The user does not have to be conscious of which sample plate container **61** the sample plate **8** is to be contained in in the plate changer **6**. The user may place the sample plate **8** in an available sample plate container **61**. An analysis process does not start with respect to a different sample plate **8** due to an erroneous operation. Even in a case where a plurality of users use a common plate changer **6**, each user can freely utilize an available sample plate container **61**.

(57) Further, in a case where the identification information of the subject sample plate **8T** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the sample plate supply control system **20** supplies the content of a batch table stored in the subject plate storage **12** to the autosampler **7**. Thus, even in a case where the sample plate **8** is contained in any sample plate container **61**, the analysis process is started with respect to the subject sample plate **8T** stored in the subject plate storage **12** based on the batch table stored in the subject plate storage **12**.

[2] Second Embodiment

(58) Next, a second embodiment of the present invention will be described. In the second embodiment, when an autosampler **7** is executing an analysis process, a plate changer **6** is in a waiting state. FIG. **7** is a flowchart showing a process executed by an identification information acquirer **13** and a supply instructor **14** of a computer **1** in the second embodiment. First, the identification information acquirer **13** of the computer **1** checks whether the identification

information of a sample plate **8** and the number of a sample plate container **61** have been acquired (step **S201**). In a case where the identification information acquirer **13** has acquired the identification information of the sample plate **8** and the number of the sample plate container **61**, the supply instructor **14** makes reference to a subject plate storage **12** (step **S202**). In a case where the identification information acquirer **13** does not acquire the identification information of the sample plate **8** and the number of the sample plate container **61**, the process ends.

(59) After the step **S202**, the supply instructor **14** determines whether the identification information of the subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13** (step **S203**). In a case where the identification information of the subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the supply instructor **14** determines that the sample plate **8** contained in the sample plate container **61** is the subject sample plate **8T** to be processed.

(60) In a case where determining that the subject sample plate **8T** is contained in the sample plate container **61**, the supply instructor **14** determines whether the autosampler **7** is executing a process with respect to another sample plate **8** (step **S204**). The supply instructor **14** acquires a processing state of the autosampler **7** by making an inquiry to a control device **4**. Alternatively, the supply instructor **14** acquires a processing state of the autosampler **7** by making an inquiry to the autosampler **7** via the control device **4**.

(61) In a case where the autosampler **7** is executing a process with respect to another sample plate **8**, the supply instructor **14** waits for a predetermined period of time (step **S205**). In a case where the predetermined period of time has elapsed, the supply instructor **14** returns to the step **S204** and determines whether the autosampler **7** is executing a process with respect to another sample plate **8**. In a case where the autosampler **7** is executing a process with respect to another sample plate **8**, the process of the step **S204** and the step **S205** is repeated.

(62) In a case where the autosampler **7** is not executing a process with respect to another sample plate **8**, the supply instructor **14** instructs the plate changer **6** to supply a sample plate **8**. The subsequent step **S206** and step **S207** will not be described since being similar to the step **S104** and the step **S105** of FIG. **6** in the first embodiment.

(63) As described above, in a case where the identification information of the subject sample plate **8T** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the sample plate supply control system **20** of the present embodiment determines whether the autosampler **7** is executing a process with respect to another sample plate **8**. When the autosampler **7** is executing a process with respect to another sample plate **8**, the sample plate supply control system **20** supplies a sample plate **8** to the autosampler **7** after the process with respect to the other sample plate **8** is completed. Thus, even when the autosampler **7** is executing a process with respect to another sample plate **8**, reservation of execution of a process with respect to a subject sample plate **8T** is made. The user can make settings for a subject sample plate **8T** regardless of whether the autosampler **7** is executing a process with respect to another sample plate **8**.

[3] Third Embodiment

(64) Next, a third embodiment of the present invention will be described. In the third embodiment, a plurality of subject sample plates **8T** are put in a standby state. As a prerequisite for the third embodiment, the identification information of a plurality of subject sample plates **8T** is stored in a subject plate storage **12**. That is, a user makes settings in regard to a plurality of plate identification numbers and selects a standby instruction button **104** with respect to a plurality of settings, in the plate setting screen **100** shown in FIG. **5**. Thus, the plurality of subject sample plates **8T** are put in the standby state.

(65) FIG. **8** is a flowchart showing a process to be executed by an identification information acquirer **13** and a supply instructor **14** of a computer **1** in the third embodiment. First, the

identification information acquirer **13** of the computer **1** checks whether the identification information of a sample plate **8** and the number of a sample plate container **61** have been acquired (step **S301**). In a case where the identification information acquirer **13** has acquired the identification information of the sample plate **8** and the number of the sample plate container **61**, the supply instructor **14** makes reference to the subject plate storage **12** (step **S302**). In a case where the identification information acquirer **13** does not acquire the identification information of the sample plate **8** and the number of the sample plate container **61**, the process ends.

(66) After the step **S302**, the supply instructor **14** determines a result of reference to the subject plate storage **12**. The supply instructor **14** determines whether the identification information of any subject sample plate **8T** among the identification information of the plurality of subject sample plates **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13** (step **S303**). In a case where the identification information of any subject sample plate **8T** stored in the subject plate storage **12** matches the identification information of the sample plate **8** acquired by the identification information acquirer **13**, the supply instructor **14** determines that the sample plate **8** contained in the sample plate container **61** is a subject sample plate **8T** to be processed.

(67) In a case where determining that the subject sample plate **8T** is contained in the sample plate container **61**, the supply instructor **14** instructs the plate changer **6** to supply the sample plate **8**. The subsequent step **S304** and step **S305** will not be described because being similar to the step **S104** and the step **S105** of FIG. **6** in the first embodiment.

(68) As described above, in a case where one of a plurality of stored identification information pieces of the subject sample plates **8T** matches the identification information of a sample plate **8** acquired by the identification information acquirer **13**, a sample plate supply control system **20** of the present embodiment supplies the sample plate **8** to the autosampler **7**. Thus, even in a case where a sample plate **8** is contained in any sample plate container **61**, any one of the plurality of subject sample plates **8T** stored in the subject plate storage **12** is supplied to the autosampler **7**. The user can make reservation for an analysis process with respect to a plurality of subject sample plates **8T** at the same time.

(69) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

[4] Correspondences Between Constituent Elements in Claims and Parts in Preferred Embodiments

(70) In the following paragraphs, non-limiting examples of correspondences between various elements recited in the claims below and those described above with respect to various preferred embodiments of the present disclosure are explained. In the above-mentioned embodiment, the plate changer **6** is an example of a plate holder, the computer **1** is an example of a terminal device, and the batch table is an example of process information.

(71) Further, in the above-mentioned embodiment, the device that functionally includes the identification information recognizer **65** of the plate changer **6** and the supply instructor **14** of the computer **1** is an example of a sample plate supply device. While one device that functionally includes the identification information recognizer **65** of the plate changer **6** and the supply instructor **14** of the computer **1** is not described by way of example in the above-mentioned embodiment, one device may include both functions.

(72) Further, in the above-mentioned embodiment, the second control program is an example of a sample plate supply control program.

(73) As each of constituent elements recited in the claims, various other elements having configurations or functions described in the claims can be also used.

[5] Other Embodiments

(74) In the above-mentioned embodiment, the plate changer **6** which is an example of a sample

plate holder is another device attached to the autosampler 7. While the sample plate holder may be another device attached to the autosampler 7, a sample plate holder may be included in the autosampler 7.

(75) While the plate changer 6 is used as a device that supplies a sample plate 8 to the autosampler 7 for a liquid chromatograph in the above-mentioned embodiment, a plate changer 6 of the present embodiment can be used as a device that supplies a sample plate 8 to an autosampler used for various analysis devices. Further, an analysis system 10 of the present invention can be used not only for a liquid chromatograph but also for another analysis device such as another type of a liquid chromatograph.

Claims

1. A sample plate supply control device that controls supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, the plurality of sample plate containers each including therein a respective contained sample plate having corresponding sample plate identification information, the sample plate supply control device comprising: a subject plate storage that stores user input identification information identifying a subject sample plate to be analyzed, in association with information of a method file relating to an analysis process; an identification information recognizer that recognizes the sample plate identification information of the contained sample plates, in association with container identification information of the sample plate containers; and a supply instructor that instructs the sample plate holder to supply the sample plate from the sample plate container to the autosampler, the sample plate container having the recognized container identification information associated with the recognized sample plate identification information that matches the sample plate identification information associated with the method file, in a case where the analysis process of the method file is performed.
2. The sample plate supply control device according to claim 1, wherein the supply instructor has a process instructor, which notifies the autosampler of process information and provides an instruction for executing the analysis process on the sample plate supplied by the sample plate holder.
3. The sample plate supply control device according to claim 1, wherein the supply instructor waits until another analysis process being executed on another plate in the autosampler is completed to instruct the sample plate holder to supply the sample plate from the one sample plate container to the autosampler.
4. The sample plate supply control device according to claim 1, further comprising a print module that prints the identification information of the subject sample plate to be attached to the subject sample plate.
5. The sample plate supply control device according to claim 1, wherein the identification information of the subject sample plate is attached to the subject sample plate as a barcode image.
6. The sample plate supply control device according to claim 1, further comprising a subject plate setter that sets the user input identification information of the subject sample plate and the method file relating to the analysis process, in association with each other, wherein the supply instructor has a process instructor that acquires the method file associated with the subject sample plate by the subject plate setter and instructs the autosampler to execute the analysis process based on the method file.
7. A sample plate supply control system comprising: an autosampler that injects a sample to be analyzed into an analysis device; a sample plate holder that has a plurality of sample plate containers, the plurality of sample plate containers each including therein a respective contained sample plate having corresponding sample plate identification information; and a terminal device that controls the sample plate holder, wherein the sample plate holder includes an identification information recognizer that recognizes the sample plate identification information of contained

sample plates, in association with the container identification information of the sample plate containers, and an identification information presenter that supplies the identification information of the contained sample plates to the terminal device, and the terminal device includes: a subject plate storage that stores user input identification information identifying a subject sample plate to be analyzed, in association with information of a method file relating to an analysis process; an identification information acquirer that acquires the identification information of the contained sample plate from the sample plate holder, and a supply instructor that instructs the sample plate holder to supply the sample plate from the sample plate container to the autosampler, the sample plate container having the recognized container identification information associated with the recognized sample plate identification information that matches the sample plate identification information associated with the method file, in a case where the analysis process of the method file is performed.

8. The sample plate supply control system according to claim 7, wherein the sample plate holder has a plurality of displays provided to correspond to the plurality of sample plate containers, and each display displays a processing state in regard to each sample plate contained in each sample plate container.

9. A sample plate supply control method of controlling supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, the plurality of sample plate containers each including therein a respective contained sample plate having corresponding sample plate identification information, the sample plate supply control method including: acquiring identification information identifying the contained sample plates from the sample plate holder, in association with container identification of the sample plate containers, wherein the identification information is input by a user and stored in a subject plate storage in association with information of a method file relating to an analysis process; instructing the sample plate holder to supply the sample plate from the sample plate container to the autosampler, the sample plate container having the recognized container identification information associated with the recognized sample plate identification information that matches the sample plate identification information associated with the method file, in a case where the analysis process of the method file is performed.

10. A non-transitory computer readable medium storing a sample plate supply control program that controls supply of a sample plate from a sample plate holder having a plurality of sample plate containers to an autosampler, the plurality of sample plate containers each including therein a respective contained sample plate having corresponding sample plate identification information, the sample plate supply control program causing a computer to execute the processes of: acquiring identification information identifying the contained sample plates from the sample plate holder, in association with container identification of the sample plate containers, wherein the identification information is input by a user and stored in a subject plate storage in association with information of a method file relating to an analysis process; instructing the sample plate holder to supply the sample plate from the sample plate container to the autosampler, the sample plate container having the recognized container identification information associated with the recognized sample plate identification information that matches the sample plate identification information associated with the method file, in a case where the analysis process of the method file is performed.
